Mobile transport layer

Snooping TCP

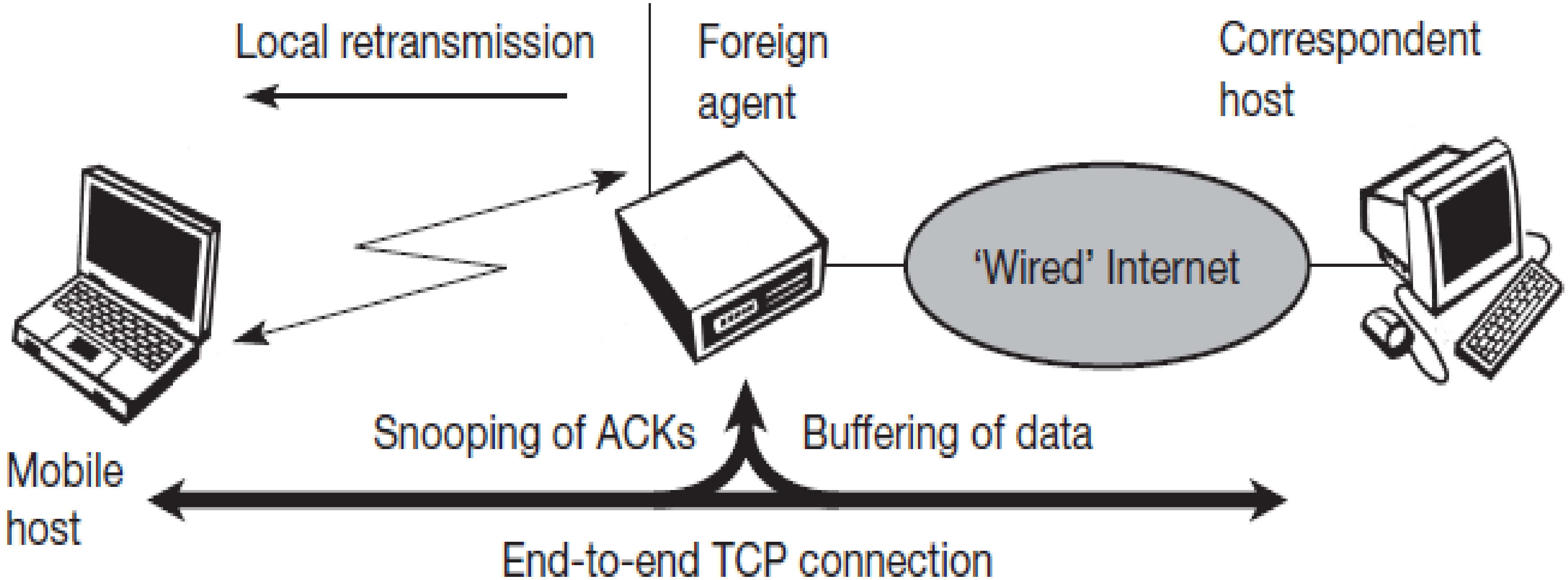
- The drawbacks of I-TCP is that it loses the original end-to-end TCP semantic.

- TCP enhancement works completely transparently and leaves the TCP end-to-end connection intact.

- The main function of the enhancement is to buffer data close to the mobile host to perform fast local retransmission in case of packet loss.

- A good place for the enhancement of TCP could be the foreign agent in the Mobile IP context.

Snooping TCP



- The foreign agent buffers all packets with **destination mobile host** and additionally 'snoops' the packet flow in both directions to recognize acknowledgements.
- The reason for buffering packets toward the mobile node is to enable the foreign agent to perform a local retransmission in case of packet loss on the wireless link.

Snooping TCP

- The foreign agent buffers every packet until it receives an ACK from the mobile host.

- If the foreign agent does not receive an ACK from the mobile host within a certain amount of time, either the packet or the ACK has been lost.

- Alternatively, the FA could receive a duplicate ACK which also shows the loss of a packet. Now the foreign agent retransmits the packet directly from the buffer.

- The time out for acknowledgements can be much shorter, because it reflects only the delay of one hop plus processing time.

Snooping TCP

- To remain transparent, the FA must not ACK data to the CN.
  - The FA can filter the duplicate ACKs to avoid unnecessary retransmissions of data from the correspondent host.

- If the foreign agent now crashes, the time-out of the correspondent host still works and triggers a retransmission.

- The foreign agent may discard duplicates of packets already retransmitted locally and acknowledged by the mobile host. This avoids unnecessary traffic on the wireless link.

Snooping TCP

- While data transfer from the mobile host with **destination correspondent host,** the foreign agent snoops into the packet stream to detect gaps in the sequence numbers of TCP.

- As soon as the foreign agent detects a missing packet, it returns a negative acknowledgement (NACK) to the mobile host.

- The mobile host can now retransmit the missing packet immediately.

- Reordering of packets is done automatically at the correspondent host by TCP.

Advantages

- The end-to-end TCP semantic is preserved. No matter at what time the FA crashes, neither the CN nor the MN have an inconsistent view of the TCP connection as is possible with I-TCP.
  - The approach automatically falls back to standard TCP if the enhancements stop working.

- The CN and MN does not need to be changed; most of the enhancements are in the foreign agent.

Advantages

- Handover of state is not required.
  - No transfer of buffer data to new FA takes place. All that happens is a time-out at the correspondent host and retransmission of the packets, possibly already to the new care-of address.

- If FA does not have this enhancement. then, the approach automatically falls back to the problems of I-TCP, since the old foreign agent may have already signaled the correct receipt of data via ACKs to the CN and now has to transfer these packets to the MN via the new FA.

Disadvantages

- Snooping TCP does not isolate the behavior of the wireless link as well as I-TCP.
  - The problems on the wireless link (interference, congestion) are now also visible for the correspondent host and not fully isolated.
  - The quality of the isolation, which snooping TCP offers, strongly depends on the quality of the wireless link, time-out values, and further traffic characteristics.
  - It is problematic that the wireless link exhibits very high delays compared to the wired link due to error correction on layer 2. This is similar to I-TCP. If this is the case, the timers in the FA and the CN are almost equal and the approach is almost ineffective.

Disadvantages

- Using negative ACKs between the foreign agent and the mobile host assumes additional mechanisms on the mobile host.
  - This approach is no longer transparent for arbitrary mobile hosts.
- Snooping and buffering data may be useless if certain encryption schemes are applied end-to-end between the CN and MN.
  - Using IP encapsulation security payload the TCP protocol header will be encrypted – snooping on the sequence numbers will no longer work.
  - Retransmitting data from the FA may be misinterpreted as replay.
  - If encryption is used above the transport layer snooping TCP can be used.

Mobile TCP

- Dropping packets due to a handover or higher bit error rates is not the only phenomenon of wireless links and mobility – the occurrence of lengthy and/or frequent disconnections is another problem.
- Quite often mobile users cannot connect at all.
- What happens to standard TCP in the case of disconnection?
- A retransmission timer that doubles with each unsuccessful ReTrans. attempt, up to a max. of 1 minute. The sender will give up after 12 ReTrans. What happens if connectivity is back earlier than this? No data is successfully transmitted for a period of 1 minute!
- The retransmission time-out is still valid and the sender has to wait. The sender also goes into slow-start because it assumes congestion.

Mobile TCP

- What happens in the case of I-TCP if the mobile is disconnected?
  - so the longer the period of disconnection, the more buffer is needed. If a handover follows the disconnection, which is typical, even more state has to be transferred to the new proxy.
- The snooping approach also suffers from being disconnected. The mobile will not be able to send ACKs so, snooping cannot help in this situation.

Mobile TCP

- The **M-TCP** approach has the same goals as I-TCP and snooping TCP: to prevent the sender window from shrinking if bit errors or disconnection but not congestion cause current problems.

- M-TCP wants to improve overall throughput, to lower the delay, to maintain end-to-end semantics of TCP, and to provide a more efficient handover.

- Additionally, M-TCP is especially adapted to the problems arising from lengthy or frequent disconnections.

Mobile TCP

- M-TCP splits the TCP connection into two parts as I-TCP does.
- An unmodified TCP is used on the standard host-**supervisory host (SH)** connection, while an optimized TCP is used on the SH-MH connection.
- The SH is responsible for exchanging data between both parts similar to the proxy in I-TCP.
- The M-TCP approach assumes a relatively low bit error rate on the wireless link. Therefore, it does not perform caching/retransmission of data via the SH.
  - If a packet is lost on the wireless link, it has to be retransmitted by the original sender. This maintains the TCP end-to-end semantics.

Mobile TCP

- The SH monitors all packets sent to the MH and ACKs returned from the MH. If the SH does not receive an ACK for some time, it assumes that the MH is disconnected. It then chokes the sender by setting the sender's window size to 0.
  - Setting the window size to 0 forces the sender to go into **persistent mode**, i.e., the state of the sender will not change no matter how long the receiver is disconnected.
  - This means that the sender will not try to retransmit data. As soon as the SH (either the old SH or a new SH) detects connectivity again, it reopens the window of the sender to the old value. The sender can continue sending at full speed. This mechanism does not require changes to the sender's TCP.

Mobile TCP

- The wireless side uses an adapted TCP that can recover from packet loss much faster.
  - This modified TCP does not use slow start, thus, M-TCP needs a **bandwidth manager** to implement fair sharing over the wireless link.

Advantages

- It maintains the TCP end-to-end semantics. The SH does not send any ACK itself but forwards the ACKs from the MH.
- If the MH is disconnected, it avoids useless retransmissions, slow starts or breaking connections by simply shrinking the sender's window to 0.
- Since it does not buffer data in the SH as I-TCP does, it is not necessary to forward buffers to a new SH. Lost packets will be automatically retransmitted to the new SH.

Disadvantages

- As the SH does not act as proxy as in I-TCP, packet loss on the wireless link due to bit errors is propagated to the sender.
  - M-TCP assumes low bit error rates, which is not always a valid assumption.

- A modified TCP on the wireless link not only requires modifications to the MH protocol software but also new network elements like the bandwidth manager.